| ID | User Story | MVP / Reach | Function | Task | Owner | Priority | Est. | Actual | Status | Acceptance Criteria |
|---|---|---|---|---|---|---|---|---|---|---|
| 1.1 | As a researcher, I want to use and improve Google's neural-network quantum decoder solution, so that I can reduce errors in a quantum computer better than other state-of-the-art decoder solutions. | MVP | Set Up | Set up github repository under which out decoder solution lives | Christopher | High | 0.75 | 3 | Done | Accessible github repository in which everyone can run and edit simple Readme file |
| 1.2 | | MVP | | Set up virtual environment for decoder solution | Christopher | High | 1.5 | 3 | Done | Only when environment is active does file inside run |
| 1.3 | | MVP | | Create Docker image and container for decoder | Christopher | High | 1 | 3 | Done | Verify image and container exists using docker ps or docker container ls |
| 1.4 | | MVP | | Run test Docker containing virtualizations | Christopher | Low | 0.75 | 1 | Done | Check if helloworld python file runs only after running container |
| 1.5 | | MVP | | Study optimizer LION from Google's paper | Tzu Chen | Medium | 4 | 13 | Done | Understand what the optimizations implemented by the Google paper are |
| 1.6 | | MVP | | Study data from Google's paper | Tzu Chen | Medium | 10 | 12 | Done | Understand the data used and implemented by Google's paper |
| 1.7 | | MVP | | Installing and learning ARC | Tzu Chen | Medium | 3.5 | 3 | Done | Run example python file in the ARC successfully and get output results |
| 1.8 | | MVP | | Study transformer and neural network models | Mara | Medium | 5 | 4.5 | In Progress | Understand neural network and transformer models to the required extent |
| 1.9 | | MVP | | Study AlphaQubit papers | Arjun | Medium | 5 | 8 | In Progress | Understand the AlphaQubit implementation and performance |
| 1.10 | | MVP | Run on Simulated Dataset | Create surface code code in Stim to simulate surface code error dataset | Christopher | Medium | 4.5 | 4 | Done | Stim circuits created for d ∈ {3,5} and specified noise model(s). |
| 1.11 | | MVP | | Generate / simulate Stim datasets | Mara | Medium | 3.75 | 2 | Done | Script generates train/val/test splits with configurable shots and p values. |
| 1.12 | | MVP | | Implement analog readout wrapper | Mara | Low | 3 | 4 | Done | Wrapper accepts soft information (e.g., logits/LLRs/probabilities) and returns calibrated analog values. |
| 1.13 | | MVP | | Connect to the ARC | Arjun | High | 3.75 | 3 | Done | SSH key + MFA working; sinfo & sbatch accessible. SLURM template committed (e.g., slurm/train_surface.sh) |
| 1.14 | | MVP | | Construct embedding scheme (with vectorizing training data and tokenization) | Arjun | Medium | 7 | 1 | Done | Deterministic mapping from syndrome/analog readout to tokens/embeddings. |
| 1.15 | | MVP | | Implement proposed transformer error correction architecture with pipeline | Christopher | Medium | 5.25 | 6 | Done | Model modules implemented with clear interfaces and docstrings. End-to-end train.py and eval.py accept con |
| 1.16 | | MVP | | Train on Stim data on ARC | Arjun | Medium | 4.75 | 6 | In Progress | SLURM job(s) complete without error; logs and checkpoints saved. |
| 1.17 | | MVP | | Test and evaluate performance on Stim data | Mara | Medium | 2.5 | 4 | Done | eval.py runs across held-out splits and p-grid; outputs LER(p) table. |
| 1.18 | | MVP | | Iteratively adjust parameters and model architecture to improve performance on Stim data | Tzu Chen | Medium | 34 | 30 | In Progress | At least N (e.g., 5) controlled experiments executed (documented changes only). Ablation table added to repo |
| 1.19 | | MVP | | Preparing pretrain, valuation and test dataset | Tzu Chen | Medium | 20 | 32 | Done | |
| 1.20 | | MVP | Run on Sycamore Dataset from Paper | Adjust embedding scheme / pipeline for sycamore dataset | Christopher | High | 55 | 60 | In Progress | Loader for Sycamore data implemented; normalization/calibration documented. |
| 1.21 | | MVP | | Train on Sycamore data on ARC | Arjun | Medium | 5.25 | | To Do | SLURM job completes. Checkpoints/logs archived separately from Stim runs. |
| 1.22 | | MVP | | Test and evaluate performance on Sycamore data | Tzu Chen | Medium | 2.75 | | Stalled | LER(p) (or per-round logical failure) computed on held-out Sycamore set. |
| 1.23 | | MVP | | Iteratively adjust parameters and model architecture to improve performance on Sycamore data | Tzu Chen | Medium | 3.75 | | Stalled | At least N (e.g., 5) controlled experiments executed for Sycamore daataset specifc challenges (documented c |
| 2.1 | As a researcher, I want to apply Google's neural-network quantum decoder to other quantum codes, so that I have a generalized decoder applicable to any type of quantum code / system. | Reach | Run on Simulated Dataset | Create code other quantum codes using Stim | Mara | Low | 7.25 | | Stalled | At least two additional codes (e.g., color code, XZZX surface) implemented. |
| 2.2 | | Reach | | Generate dataset for other quantum codes | Christopher | Low | 4.5 | | Stalled | Datasets built with the same schema; metadata includes code_family. |
| 2.3 | | Reach | | Adapt readout wrapper to accept other datasets | Christopher | Low | 4.25 | | To Do | Wrapper adapts to code family differences via config (no code changes). |
| 2.4 | | Reach | | Adapt embedding scheme to accept different quantum code data (with unique vectorization schemes) | Arjun | Low | 6.25 | 8 | Done | Single embedding API supports all code families (feature flags). Validation asserts consistent tensor shapes |
| 2.5 | | Reach | | Train on Stim data for other quantum codes | Christopher | Low | 5.25 | | Stalled | Training runs complete for each additional code (min one config each). |
| 2.6 | | Reach | | Test and evaluate performance on Stim data for other quantum codes | Christopher | Low | 5.5 | | Stalled | LER(p) curves produced and saved for each code. |
| 3.1 | As a graduate student, I would like to compare my decoder with a pre-built transformer-based decoder and baseline models | MVP | Implement baselines | Implement MWPM baseline error correction on simulated Stim data | Arjun | Medium | 3.75 | 2 | Done | MWPM decoder (e.g., PyMatching) integrated with same data interface. |
| 3.2 | | MVP | | Evaluate MWPM baseline performance on Stim data | Tzu Chen | Medium | 2.5 | 2 | Done | LER(p) for MWPM on surface code computed and saved. |
| 3.3 | | MVP | | Implement MWPM baseline error correction on Sycamore data | Tzu Chen | Medium | 4 | | Stalled | Sycamore loader into MWPM path runs end-to-end. |
| 3.4 | | MVP | | Evaluate MWPM baseline performance on Sycamore data | Mara | High | 3.5 | | Stalled | LER results computed; plot and results exported. |
| 3.5 | | MVP | Implement Gogle's transformer model | Build branch from neural net pipeline to send embedded data to Google's transformer model | Tzu Chen | Low | 4.25 | 1 | Done | Separate training path --model google_transformer implemented. |
| 3.6 | | MVP | | Build transformer based on Google's pseudo code | Tzu Chen | Medium | 8 | 10 | Done | Verify it uilizies same parameters, architectural design, is able to handle similar error data, and performs at lea |
| 3.7 | | MVP | | Run transformer based on Google's pseudo code over ARC | Arjun | Medium | 5 | 8 | Done | Verify a run can be made over the provided allocation over ARC usingSTIM generated data and model based o |
| 3.8 | | MVP | | Train Google's transformer model on Stim Data | Christopher | Medium | 5 | | To Do | Training completes with logged metrics and checkpoints. |
| 3.9 | | MVP | | Evaluate Google's transformer performance on Stim data | Tzu Chen | Low | 2.75 | | To Do | LER(p) computed and saved. Plots exported. |
| 3.10 | | MVP | | Train Google's transformer model on Sycamore Data | Arjun | Medium | 5 | | Stalled | SLURM job completes. Artifacts saved. |
| 3.11 | | MVP | | Evaluate Google's transformer performance on Sycamore data | Christopher | High | 3.25 | | Stalled | Evaluation artifacts produced. Export plots and results. |
| 3.12 | | MVP | Evaluation Module | Compare performance with proposed solution using LER (p) evaluation metric | Arjun | High | 3.5 | | To Do | - Unified comparison plot with all decoders (ours, Google's, MWPM) for Stim & Sycamore.<br>- Table reports LER at each p, relative improvement vs MWPM, and crossover points.<br>- AUC (LER vs p) computed; bootstrap CIs provided.<br>- Narrative summary added to report. |
| 3.13 | | MVP | | Generate visualizations and report | Mara | Medium | 30 | 29.5 | In Progress | Figures: LER(p), training curves, ablations, throughput/latency, calibration. |
| 4.1 | As a graduate student, I want to integrate a neural-network quantum decoder solution into a simplified quantum model. | Reach | Decoder Adoption | Adapt readout module | Arjun | Low | 2.75 | | Stalled | Streaming input (e.g., ZeroMQ/Kafka/pipe) interface implemented. |
| 4.2 | | Reach | | Feed into pipeline of transformer model that evaluates error correction output | Tzu Chen | Low | 3.5 | 6 | Done | End-to-end streaming decode runs continuously ≥30 minutes without crash. |
| 4.3 | | Reach | | Evaluate and verify real-time performance and throughput | Tzu Chen | Low | 4 | | Stalled | Latency (p50/p95) and throughput measured at batch=1 and batch=256. |
| 4.4 | | Reach | | Research adapting decoder to other quantum code types | Arjun | Low | 10 | 6 | In Progress | Finalize possible adaptation documentation with steps to be taken, concerns, and future work |
| 4.5 | | MVP | Interface and packaging | Construct interface to provide option to modify parametrs of | Mara | Medium | 4.25 | | Stalled | CLI + minimal UI (e.g., Streamlit or REST) exposes key params (depth, heads, lr, batch, calibration). Validation |
| 4.6 | | MVP | | Build documentation and details on model functionality and | Arjun | Medium | 12 | 8 | In Progress | Verify correctness with Alphaqubit paper and ensure model data and details are accurate |